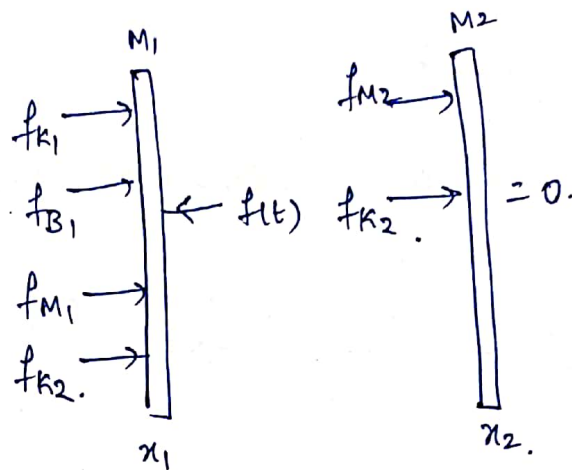
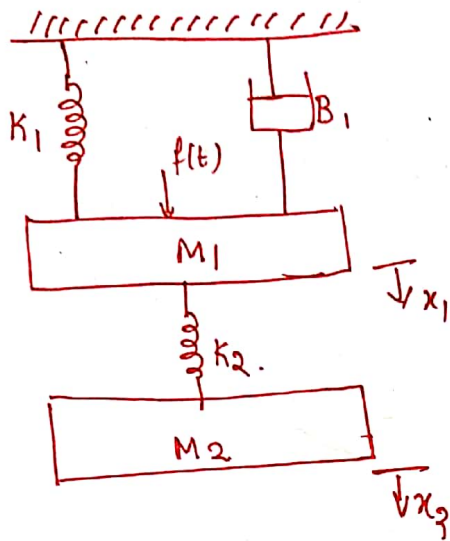


Transfer function (Tutorial 2)

①

Problems (Class B)

①



for x_1

$$M_1 \ddot{x}_1 + B_1 \dot{x}_1 + K_1 x_1 + K_2 (x_1 - x_2) = f(t) \rightarrow \textcircled{1}$$

$$M_2 \ddot{x}_2 + K_2 (x_2 - x_1) = 0 \rightarrow \textcircled{2}$$

taking LT.

$$\textcircled{1} \Rightarrow M_1 s^2 x_1(s) + B_1 s x_1(s) + K_1 x_1(s) + K_2 [x_1(s) - x_2(s)] = F(s).$$

$$\boxed{[M_1 s^2 + B_1 s + (K_1 + K_2)] x_1(s) - K_2 x_2(s) = F(s)} \rightarrow \textcircled{a}$$

$$\textcircled{2} \Rightarrow M_2 s^2 x_2(s) + K_2 x_2(s) - K_2 x_1(s) = 0$$

$$\boxed{-K_2 x_1(s) + [M_2 s^2 + K_2] x_2(s) = 0} \rightarrow \textcircled{b}$$

$$\begin{bmatrix} M_1 s^2 + B_1 s + K_1 + K_2 & -K_2 \\ -K_2 & (M_2 s^2 + K_2) \end{bmatrix} \begin{bmatrix} x_1(s) \\ x_2(s) \end{bmatrix} = \begin{bmatrix} F(s) \\ 0 \end{bmatrix}$$

we should find TF $\frac{x_2(s)}{F(s)}$, $\therefore$ By Cramer's rule.

$$X_2(s) = \frac{\begin{vmatrix} M_1 s^2 + b_1 s + k_1 + k_2 & F(s) \\ -k_2 & 0 \end{vmatrix}}{\Delta}$$

where $\Delta = (M_1 s^2 + b_1 s + k_1 + k_2)(M_2 s^2 + k_2) - k_2^2$

then

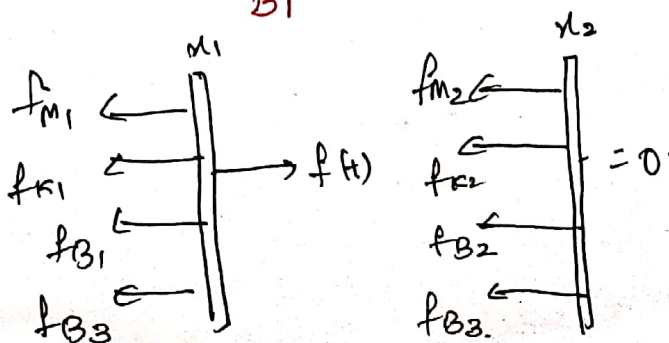
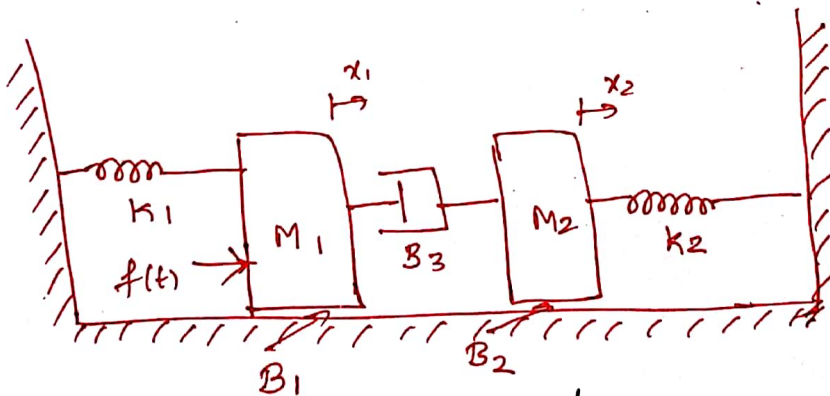
$$X_2(s) = \frac{k_2 F(s)}{\Delta} \Rightarrow \boxed{\frac{X_2(s)}{F(s)} = \frac{k_2}{\Delta}}$$

we should find $TR = X_1(s)/F(s)$ then by Cramer's Rule

$$X_1(s) = \frac{\begin{vmatrix} F(s) & -k_2 \\ 0 & (M_2 s^2 + k_2) \end{vmatrix}}{\Delta}$$

$$\Rightarrow \boxed{\frac{X_1(s)}{F(s)} = \frac{M_2 s^2 + k_2}{\Delta}}$$

(2)



(2)

$$M_1 \ddot{x}_1 + k_1 x_1 + B_1 \dot{x}_1 + B_3 (\dot{x}_1 - \dot{x}_2) = f(t) \rightarrow (1)$$

$$M_2 \ddot{x}_2 + B_2 \dot{x}_2 + k_2 x_2 + B_3 (\dot{x}_2 - \dot{x}_1) = 0 \rightarrow (2)$$

Apply LT

$$(1) \Rightarrow M_1 s^2 x_1(s) + k_1 x_1(s) + B_1 s x_1(s) + B_3 s (x_1(s) - x_2(s)) = F(s)$$

$$\boxed{[M_1 s^2 + (B_1 + B_3)s + k_1] x_1(s) - B_3 s x_2(s) = F(s)} \rightarrow (a)$$

$$(2) \Rightarrow M_2 s^2 x_2(s) + B_2 s x_2(s) + k_2 x_2(s) + B_3 s (x_2(s) - x_1(s)) = 0$$

$$\boxed{[M_2 s^2 + (B_2 + B_3)s + k_2] x_2(s) - B_3 s x_1(s) = 0} \rightarrow (b)$$

Putting in matrix form

$$\begin{bmatrix} M_1 s^2 + (B_1 + B_3)s + k_1 & -B_3 s \\ -B_3 s & M_2 s^2 + (B_2 + B_3)s + k_2 \end{bmatrix} \begin{bmatrix} x_1(s) \\ x_2(s) \end{bmatrix} = \begin{bmatrix} F(s) \\ 0 \end{bmatrix}$$

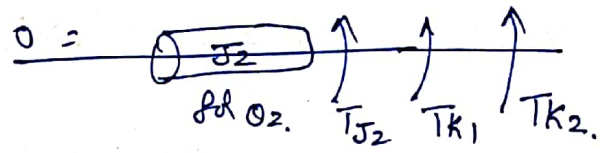
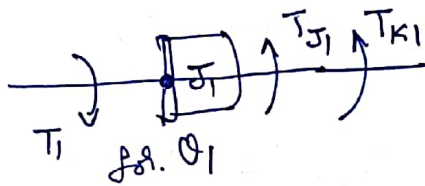
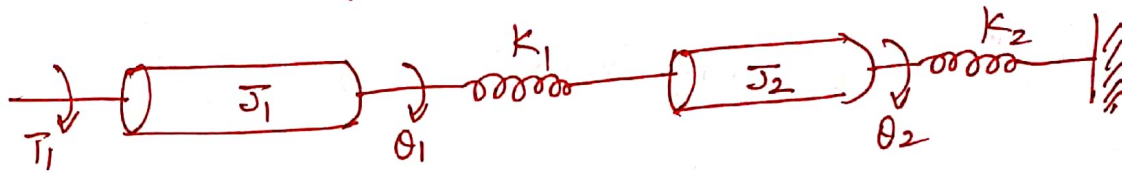
$$\text{let } \Delta = [M_1 s^2 + (B_1 + B_3)s + k_1][M_2 s^2 + (B_2 + B_3)s + k_2] - B_3^2 s^2$$

$$\text{then } x_1(s) = \frac{\begin{vmatrix} F(s) & -B_3 s \\ 0 & M_2 s^2 + (B_2 + B_3)s + k_2 \end{vmatrix}}{\Delta}$$

$$\& \frac{x_1(s)}{F(s)} = \frac{M_2 s^2 + (B_2 + B_3)s + k_2}{\Delta}$$

$$\& \frac{x_2(s)}{F(s)} = \frac{B_3 s}{\Delta}$$

③ Rotational system



At node θ_1

$$J_1 \ddot{\theta}_1 + K_1 (\theta_1 - \theta_2) = T_1 \longrightarrow \textcircled{1}$$

At node θ_2

$$J_2 \ddot{\theta}_2 + K_2 \theta_2 + K_1 (\theta_2 - \theta_1) = 0 \longrightarrow \textcircled{2}$$

take Laplace.

$$\textcircled{1} \Rightarrow J_1 s^2 \theta_1(s) + K_1 (\theta_1(s) - \theta_2(s)) = T_1(s)$$

$$\boxed{[J_1 s^2 + K_1] \theta_1(s) - K_1 \theta_2(s) = T_1(s)} \longrightarrow \textcircled{a}$$

$$\textcircled{2} \quad J_2 s^2 \theta_2(s) + K_2 \theta_2(s) + K_1 (\theta_2(s) - \theta_1(s)) = 0$$

$$\boxed{-K_1 \theta_1(s) + [J_2 s^2 + (K_1 + K_2)] \theta_2(s) = 0} \longrightarrow \textcircled{b}$$

matrix form

$$\begin{bmatrix} J_1 s^2 + K_1 & -K_1 \\ -K_1 & J_2 s^2 + (K_1 + K_2) \end{bmatrix} \begin{bmatrix} \theta_1(s) \\ \theta_2(s) \end{bmatrix} = \begin{bmatrix} T_1(s) \\ 0 \end{bmatrix}$$

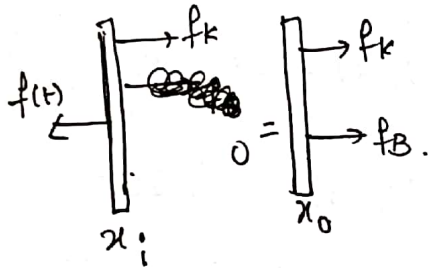
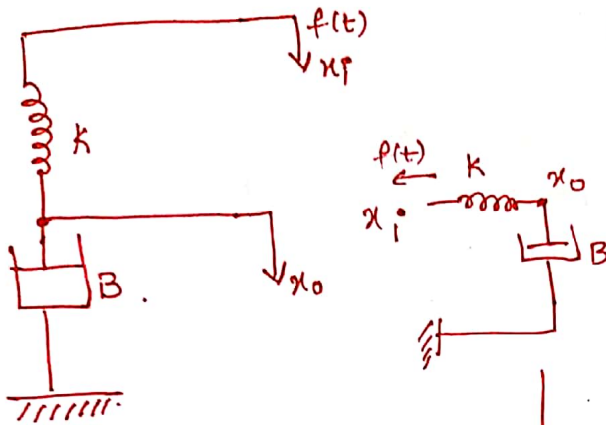
$$\text{let } \Delta = (J_1 s^2 + K_1)(J_2 s^2 + (K_1 + K_2)) - K_1^2$$

then

$$\boxed{\frac{\theta_1(s)}{T_1(s)} = \frac{J_2 s^2 + K_1 + K_2}{\Delta}} \quad , \quad \boxed{\frac{\theta_2(s)}{T_1(s)} = \frac{K_1}{\Delta}}$$

Q. Write differential equations for the given two systems and find their TF's & thereby P.T they are analogous.

Q. ①



for x_o

$$B \cdot \frac{dx_o}{dt} + k(x_o - x_i) = 0 \rightarrow ①$$

for x_i

$$k(x_i - x_o) = f(t) \rightarrow ②$$

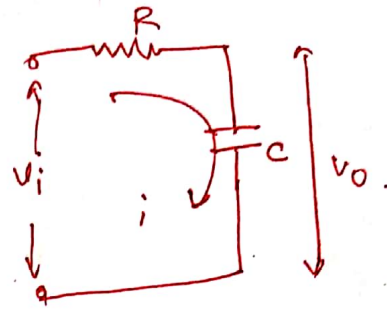
take LP

$$① \Rightarrow B \cdot s \cdot x_o(s) + k x_o(s) - k \cdot x_i(s) = 0$$

$$(Bs + k) x_o(s) - k \cdot x_i(s) = 0$$

$$\boxed{\frac{x_o(s)}{x_i(s)} = \frac{k}{Bs + k} = \frac{1}{1 + \left(\frac{B}{k}\right)s}} \rightarrow ①$$

②



$$V_i(s) = \left(R + \frac{1}{Cs}\right) I(s)$$

$$V_o(s) = \frac{1}{Cs} \cdot I(s)$$

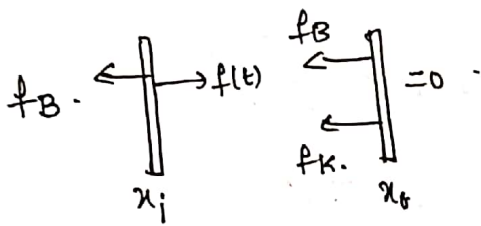
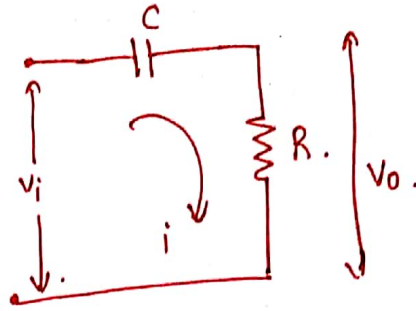
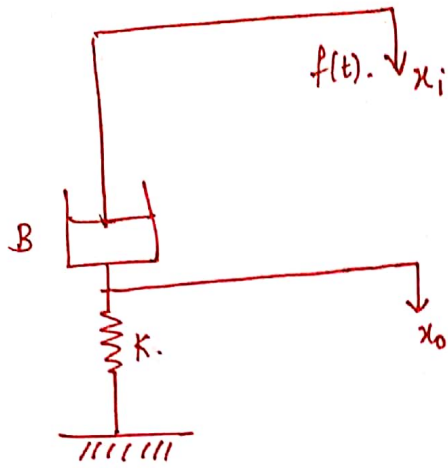
$$\frac{V_o(s)}{V_i(s)} = \frac{1/Cs}{R + \frac{1}{Cs}}$$

$$\boxed{\frac{V_o(s)}{V_i(s)} = \frac{1}{1 + (RC)s}} \rightarrow ②$$

Comparing Eqn ① & ② we find that they are analogous.

$$\therefore B = R \text{ \& } \frac{1}{k} = C$$

4(b)



$$V_i(s) = I(s) \left(R + \frac{1}{Cs} \right)$$

$$V_o(s) = R \cdot I(s)$$

$$B[\dot{x}_i - \dot{x}_o] = f(t) \rightarrow (1)$$

$$B[\dot{x}_o - \dot{x}_i] + K \cdot x_o = 0 \rightarrow (2)$$

$$B s [x_o(s) - x_i(s)] + K \cdot x_o(s) = 0$$

$$\frac{x_o(s)}{x_i(s)} = \frac{Bs}{Bs + K} = \frac{(B/K)s}{1 + (B/K)s}$$

→ (a)

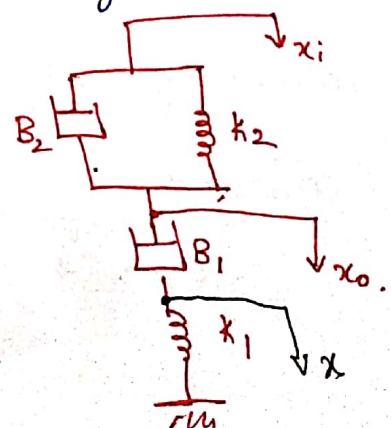
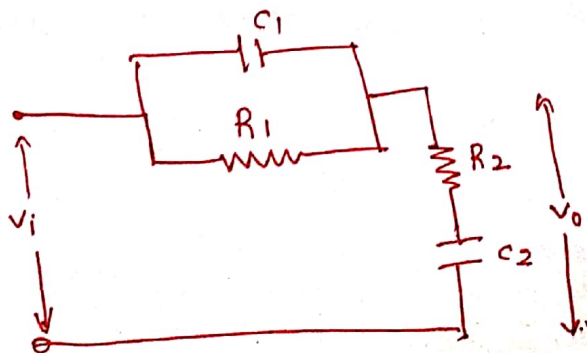
$$\frac{V_o(s)}{V_i(s)} = \frac{R}{R + \frac{1}{Cs}} = \frac{(RC)s}{(RC)s + 1} \rightarrow (b)$$

comparing (a) & (b)

$$B = R, \quad K = \frac{1}{C} \text{ or } C = \frac{1}{K}$$

The above two Equations are analogous.

4(c)



C_1 is parallel with R_1

$$Z_1 = \frac{R_1 \cdot \frac{1}{C_1 s}}{R_1 + \frac{1}{C_1 s}} = \frac{R_1}{1 + R_1 C_1 s}$$

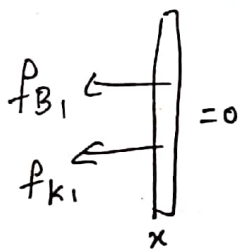
R_2 in series with C_2

$$Z_2 = R_2 + \frac{1}{C_2 s} = \frac{R_2 C_2 s + 1}{C_2 s}$$

$$\frac{V_o(s)}{V_i(s)} = \frac{Z_2}{Z_1 + Z_2} = \frac{\frac{R_2 C_2 s + 1}{C_2 s}}{\frac{R_1}{1 + R_1 C_1 s} + \frac{R_2 C_2 s + 1}{C_2 s}} = \frac{\frac{(R_2 C_2 s + 1)}{C_2 s}}{\frac{R_1 C_2 s + R_2 C_2 s + 1}{(1 + R_1 C_1 s) C_2 s}}$$

$$\boxed{\frac{V_o(s)}{V_i(s)} = \frac{(1 + R_2 C_2 s)(1 + R_1 C_1 s)}{R_1 C_2 s + (1 + R_2 C_2 s)(1 + R_1 C_1 s)}} \rightarrow \textcircled{1}$$

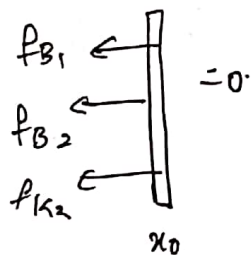
For mechanical system



$$K_1 x + B_1 (\dot{x} - \dot{x}_0) = 0$$

$$K_1 x(s) + B_1 s (x(s) - x_0(s)) = 0$$

$$\Rightarrow \boxed{x(s) = \left[\frac{B_1 s}{K_1 + B_1 s} \right] x_0(s)}$$



$$B_1 [\dot{x}_0 - \dot{x}_i] + K_2 [x_0 - x_i] + B_2 [\dot{x}_0 - \dot{x}_i] = 0$$

$$B_1 s [x_0(s) - x_i(s)] + K_2 [x_0(s) - x_i(s)] + B_2 s [x_0(s) - x_i(s)] = 0$$

$$B_1 s \left[x_0(s) - \left[\frac{B_1 s}{K_1 + B_1 s} \right] x_0(s) \right] + K_2 [x_0(s) - x_i(s)] + B_2 s [x_0(s) - x_i(s)] = 0$$

$$x_0(s) \left[B_1 (s) - \frac{B_1 s}{K_1 + B_1 s} + K_2 + B_2 s \right] = [K_2 + B_2 s] x_i(s)$$

$$\frac{X_0(s)}{X_i(s)} = \frac{(K_2 + B_2 s)(K_1 + B_1 s)}{B_1 K_1 s + \cancel{B_1^2 s^2} - \cancel{B_2^2 s^2} + K_1 K_2 + K_2 B_1 s + K_1 B_2 s + B_1 B_2 s^2}$$

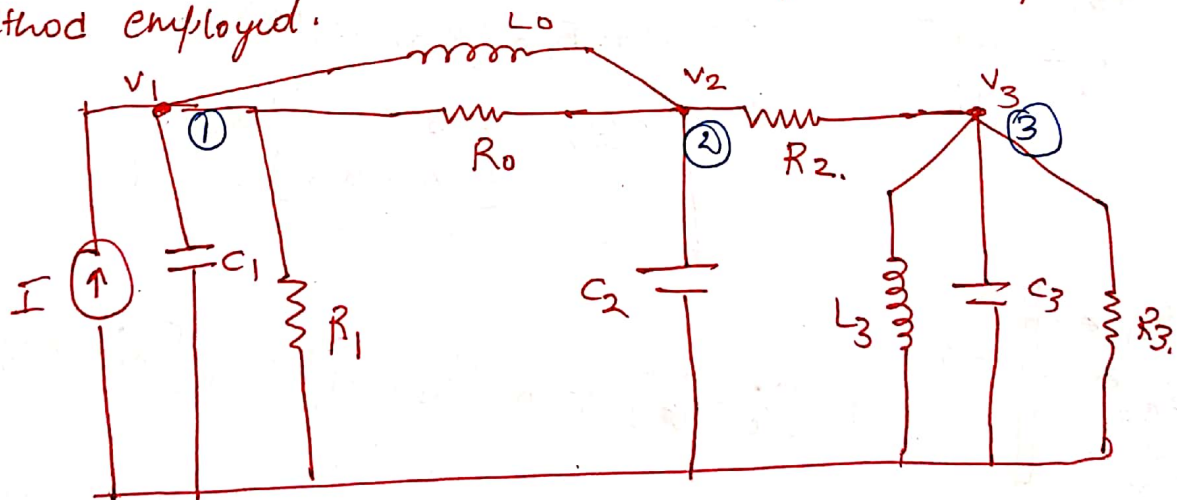
$$\frac{X_0(s)}{X_i(s)} = \frac{K_1 K_2 \left(1 + \frac{B_2 s}{K_2}\right) \left(1 + \frac{B_1 s}{K_1}\right)}{K_1 K_2 \left[\frac{B_1}{K_2} s + 1 + \frac{B_1}{K_1} s + \frac{B_2 s}{K_2} + \frac{B_1 B_2 s^2}{K_1 K_2} \right]}$$

on simplifying.

$$\frac{X_0(s)}{X_i(s)} = \frac{\left(\frac{B_2 s}{K_2} + 1\right) \left(1 + \frac{B_1}{K_1} s\right)}{\frac{B_1}{K_2} s + \left(\frac{B_2 s}{K_2} + 1\right) \left(\frac{B_1}{K_1} + 1\right)}$$

$$R_2 = B_2 \quad C_2 = \frac{1}{K_2} \quad R_1 = B_1$$

⑤ Figure shows. FI analogous electric ckt of a translational mechanical system, Draw the mechanical system and explain the method employed.



Note
=

F	i
M	c
B	G
K	$\frac{1}{L}$
x	ψ
Velocity	voltage V

From KCL

for node ①

$$I = C_1 \frac{dv_1}{dt} + G_1 v_1 + G_0 (v_1 - v_2) + \frac{1}{L_0} \int (v_1 - v_2) dt$$

→ ①

∴ FADFI FI analogy

(5)

$$F = M_1 \frac{d^2 x_1}{dt^2} + B_1 \frac{dx_1}{dt} + B_0 \left(\frac{dx_1}{dt} - \frac{dx_2}{dt} \right) + k_0 (x_1 - x_2) \Rightarrow \textcircled{a}$$

KCL
for node (2)

$$G_0 (v_2 - v_1) + C_2 \frac{dv_2}{dt} + G_2 (v_2 - v_3) + \frac{1}{L_0} \int (v_2 - v_1) \cdot dt = 0 \rightarrow \textcircled{2}$$

From FI analogy

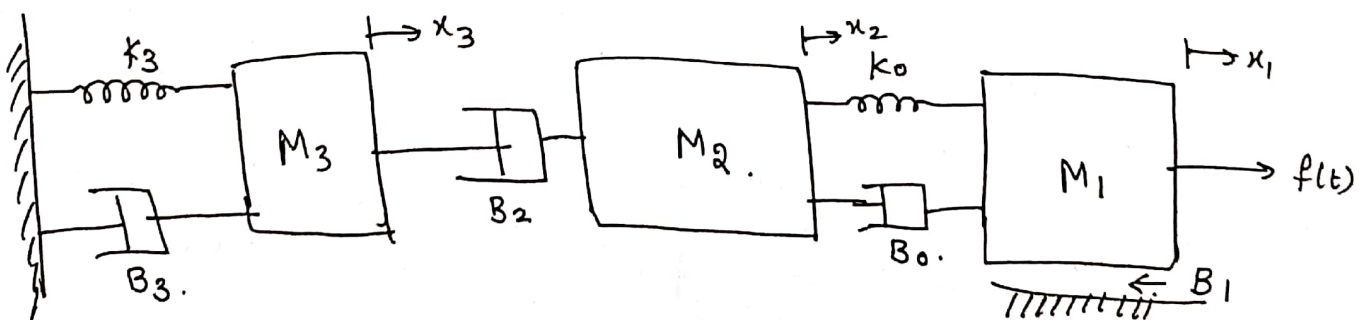
$$B_0 (\dot{x}_2 - \dot{x}_1) + M_2 \frac{d^2 x_2}{dt^2} + B_2 (\dot{x}_2 - \dot{x}_3) + k_0 (x_2 - x_3) = 0 \rightarrow \textcircled{b}$$

KCL
for node (3)

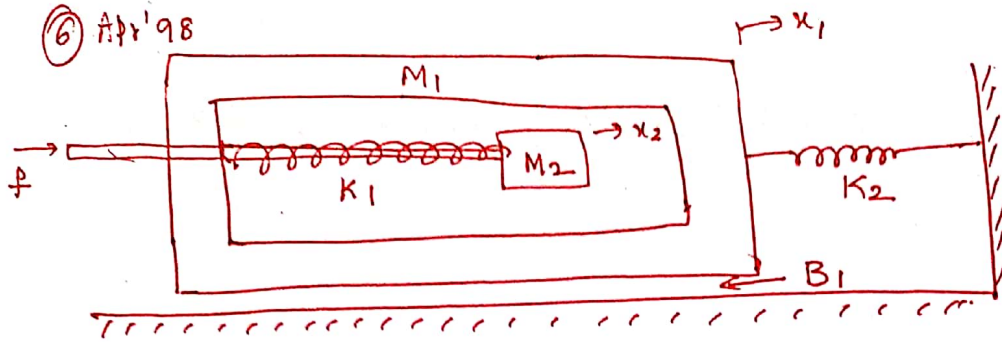
$$G_2 (v_3 - v_2) + \frac{1}{L_3} \int v_3 \cdot dt + C_3 \frac{dv_3}{dt} + G_3 v_3 = 0 \rightarrow \textcircled{3}$$

from FI - analogy

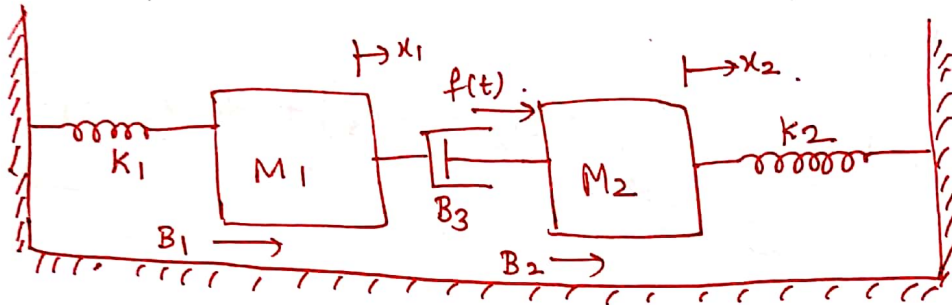
$$B_2 (\dot{x}_3 - \dot{x}_2) + k_3 \cdot x_3 + M_3 \frac{d^2 x_3}{dt^2} + B_3 \dot{x}_3 = 0 \rightarrow \textcircled{c}$$



⑥ Apr '98



⑦ Aug '91



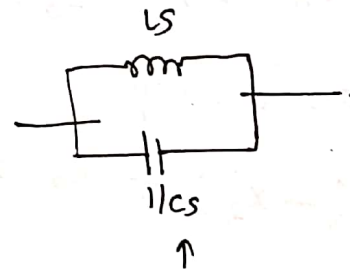
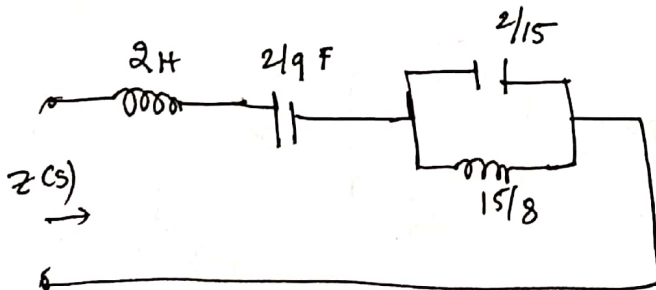
⑧

$$Z(s) = \frac{2(s^2+1)(s^2+9)}{s(s^2+4)}$$

Expanding by PF method.

$$Z(s) = 2s + \frac{9/2}{s} + \frac{15/2 s}{s^2+4}$$

$$4c = 15/2$$



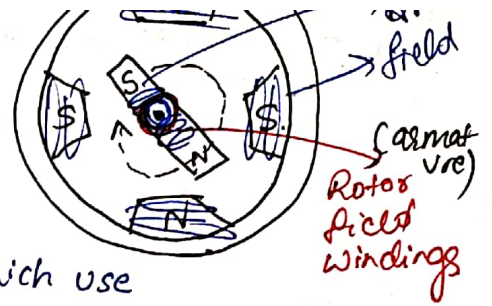
$$Z_p = \frac{Ls \cdot \left(\frac{1}{cs}\right)}{Ls + \frac{1}{cs}}$$

$$Z_p = \frac{L/c}{Lcs^2 + 1}$$

$$Z_p = \frac{4c}{\left(s^2 + \frac{1}{Lc}\right)Lc}$$

Eg: Dc motor, AC motor (Actuators)

DC Motor (consists of Rotating part $\rightarrow$ Rotor & Stationary part $\rightarrow$ Stator)



DC motors are electromechanical devices, which use interaction of magnetic fields and conductors to convert electrical energy into rotary mechanical energy.

(can be used to rotate pumps, fans, wheels etc)

DC motor equivalent circuit

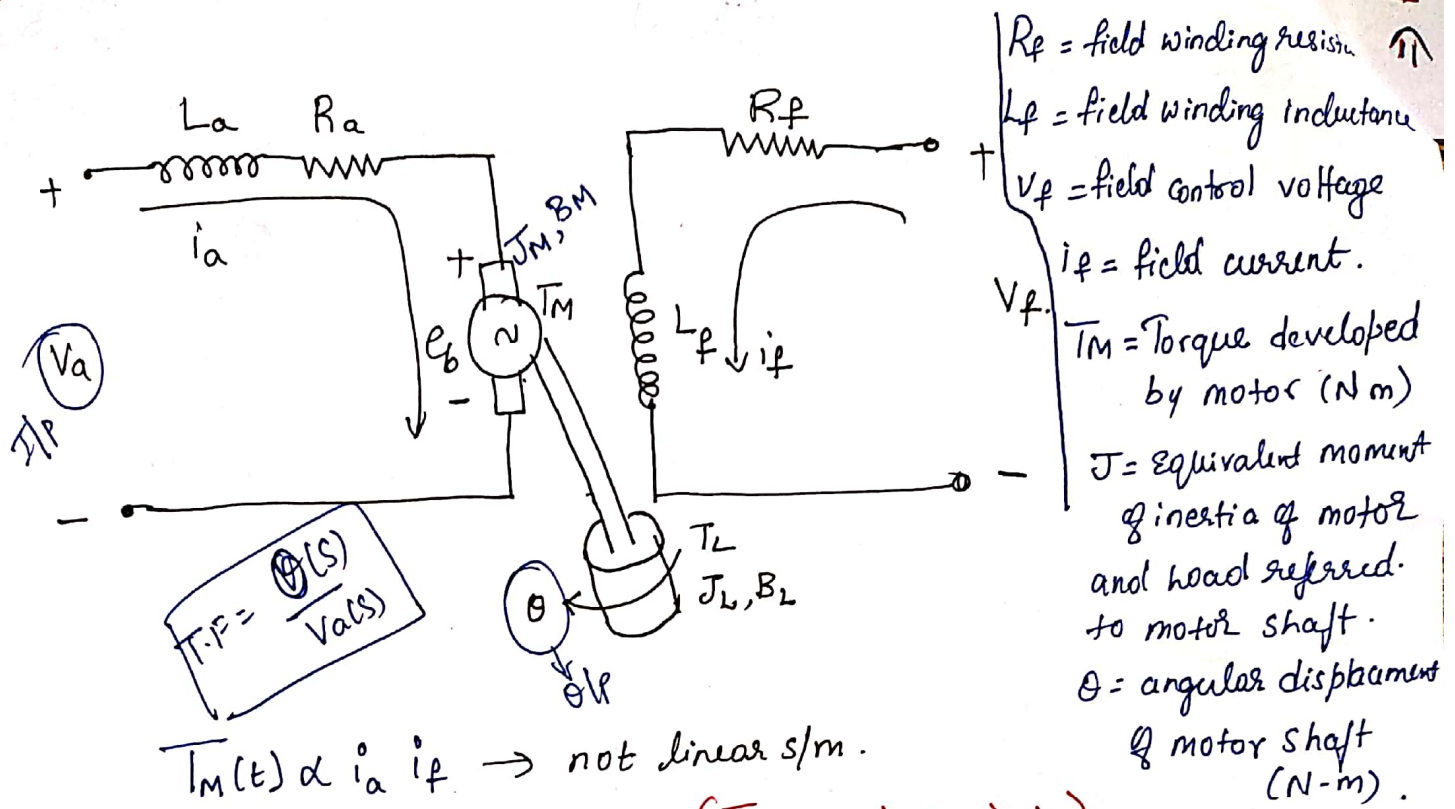
DC motor has two distinct circuit.

① field circuit

② Armature circuit

Input is electrical power & o/p is mechanical power

- ① In this equivalent circuit, the field winding is supplied from a separate DC voltage source V_f
- ② R_f & L_f represent the resistance and inductance of the field winding.
- ③ The current I_f produced in winding establishes the magnetic field necessary for motor operation.
- ④ In armature circuit (rotor), V_T is the voltage applied across the motor terminals,
- ⑤ I_a is the current flowing in the armature ckt, R_a is the resistance of armature winding, L_a is the inductance of armature winding.
- ⑥ E_b is the total voltage induced in the armature (Back EMF) when armature of a DC motor rotates, under the influence of driving torque, the armature conductors move through the magnetic field & hence emf is induced that acts in opposite direction to the applied voltage V .



$$T_M(t) \propto i_a i_f \rightarrow \text{not linear s/m.}$$

How do we make it linear? ($T_M(t) = k_1 k_f i_f i_a$).

$\rightarrow$ fix i_a $i_f \downarrow \uparrow \Rightarrow$ field controlled $\Rightarrow$ fix i_a & vary i_f .

$\rightarrow$ fix i_f $i_a \downarrow \uparrow \Rightarrow$ armature controlled $\Rightarrow$ fix i_f & vary i_a

$\Rightarrow$ Suppose i_f is a constant then

$$T_M(t) = k_1 i_a(t).$$

(Armature controlled DC motor)

$$\text{Back emf } e_b = k_2 \cdot \frac{d\theta}{dt} \rightarrow \text{angular velocity.}$$

$\uparrow$
 occurs when there is relative motion b/w rotor & stator.
 E is generated inside the coil.

To find $i_a(t)$, let us take dynamics of LHS.

$$V_a(t) = L_a \cdot \frac{di_a}{dt} + R_a i_a + e_b(t)$$

$$T(t) = J_m \cdot \frac{d^2\theta}{dt^2} + B_m \frac{d\theta}{dt} + T_L(t) \rightarrow \text{load torque.} \quad (4)$$

$$\text{where } T_L(t) = J_L \frac{d^2\theta}{dt^2} + B_L \frac{d\theta}{dt} \rightarrow (5)$$

⇒ How do we find TF?

(2)

$$i/p = V_a, \quad o/p = \theta$$

$\searrow$ cause $\searrow$ effect

⇒ Under zero initial conditions.

$$V_a(s) = L_a s I_a(s) + R_a I_a(s) + \overbrace{k_2 s}^{E_b(s)} \theta(s)$$

$$\boxed{V_a(s) = (L_a s + R_a) I_a(s) + k_2 s \theta(s)} \rightarrow (1)$$

$$T(s) = J_m s^2 \theta(s) + B_m s \theta(s) + J_L s^2 \theta(s) + B_L s \theta(s)$$

$$\boxed{T(s) = [(J_m + J_L) s^2 + (B_m + B_L) s] \theta(s)} \rightarrow (2)$$

where $T(s) = k_1 I_a(s) \Rightarrow \boxed{I_a(s) = \frac{T(s)}{k_1}} \rightarrow (3)$

$$(1) \Rightarrow V_a(s) = (L_a s + R_a) \frac{T(s)}{k_1} + k_2 s \theta(s)$$

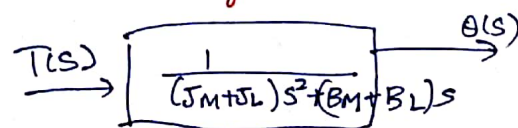
$$(2) \Rightarrow V_a(s) = \frac{(L_a s + R_a) [(J_m + J_L) s^2 + (B_m + B_L) s] + k_2 s}{k_1} \theta(s)$$

$$\frac{\theta(s)}{V_a(s)} = \frac{k_1}{(L_a s + R_a) [(J_m + J_L) s^2 + (B_m + B_L) s] + k_1 k_2 s}$$

$$\frac{\theta(s)}{V_a(s)} = \frac{1}{s} \cdot \frac{k_1}{(L_a s + R_a) [(J_m + J_L) s + (B_m + B_L)] + k_1 k_2}$$

⇒ 3rd order system.

⇒ Armature controlled DC motor acts like integrator because there is pole at origin.

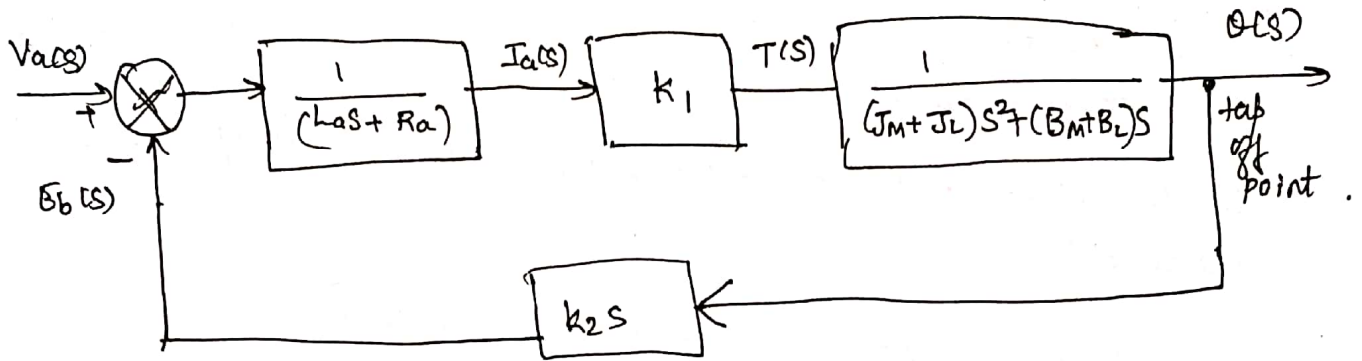


$$\Rightarrow \text{from (2)} \quad \frac{\theta(s)}{T(s)} = \frac{1}{(J_m + J_L) s^2 + (B_m + B_L) s}$$

$$\Rightarrow \text{from (3)} \quad T(s) | I_a(s) = k_1 \quad I_a(s) \rightarrow \boxed{k_1} \rightarrow T(s)$$

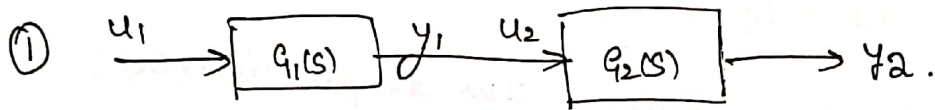
(2)

$$(iii) V_a(s) - E_b(s) = (Ls + R_a) I_a(s)$$



⇒ Signals from L-R ⇒ forward path .

⇒ Signals from tap-off point to h ⇒ feedback path .

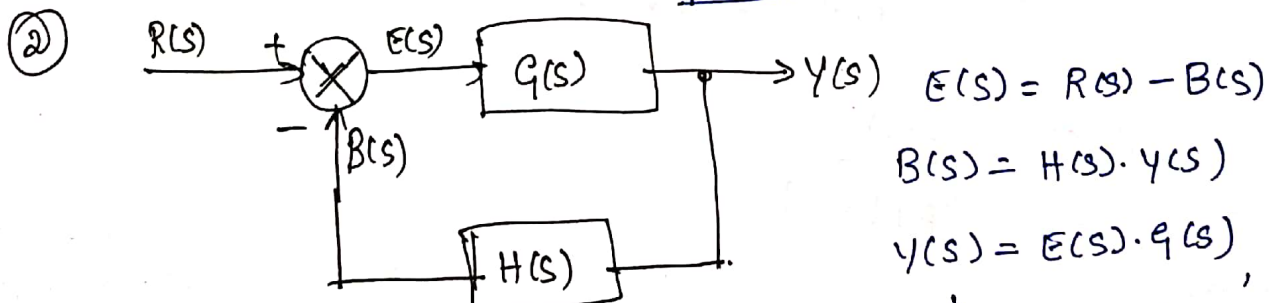
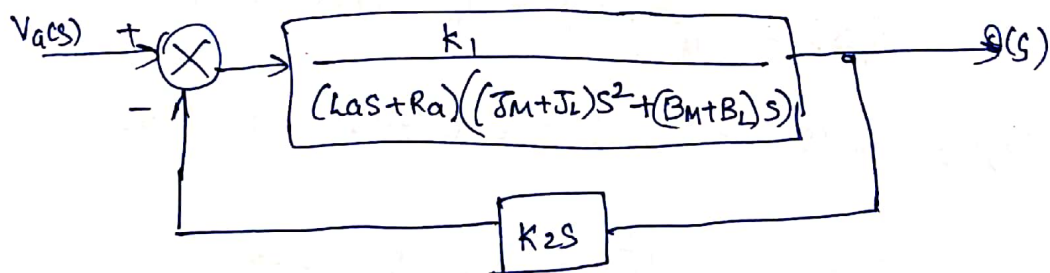
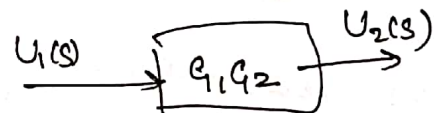


$$\frac{y_2(s)}{u_2(s)} = G_2(s) \Rightarrow y_2(s) = G_2(s) u_2(s)$$

$$\frac{y_1(s)}{u_1(s)} = G_1(s) \Rightarrow y_1(s) = G_1(s) u_1(s)$$

$$y_1(s) = u_2(s)$$

$$y_2(s) = G_2(s) G_1(s) u_1(s)$$



$$E(s) = R(s) - B(s)$$

$$B(s) = H(s) \cdot Y(s)$$

$$Y(s) = E(s) \cdot G(s)$$

$$\therefore E(s) = R(s) - H(s) \cdot E(s) \cdot G(s)$$

$$R(s) = E(s) + H(s) \cdot G(s) \cdot E(s)$$

$$R(s) = E(s) (1 + G(s) \cdot H(s))$$

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G(s) \cdot H(s)}$$

$$\frac{Y(s)}{R(s) \cdot G(s)} = \frac{1}{1 + G(s) \cdot H(s)}$$

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s) \cdot H(s)}$$

Field controlled DC motor

$$T_m(t) = (k_i k_f i_a) i_f(t)$$

$$T_m(t) = k_m i_f(t) \rightarrow \textcircled{3}$$

where $k_m = k_i k_f i_a$ $T_m(s) = k_m I_f(s)$
Torque developed by the motor is now linearly proportional to current (i_f)

To find $i_f(t)$, let us take dynamics of RHS.

$$L_f \cdot \frac{di_f(t)}{dt} + R_f i_f = V_f(t) \text{ take laplace}$$

$$\therefore V_f(s) = [R_f + L_f \cdot s] I_f(s) \rightarrow \textcircled{4}$$

The motor torque (T_m) is equal to the torque delivered to the load

$$T_m(s) = T_L(s) + T_d(s) \rightarrow \textcircled{5}$$

where T_L = Load torque

T_d = Disturbance torque (due to external forces like wind etc)

The load torque is given by

$$T_L(s) = J_m s^2 \theta(s) + B_m s \cdot \theta(s) + T_d(s)$$

$$T_L(s) = T_m(s) - T_d(s) = J_m s^2 \theta(s) + B_m s \theta(s) + T_d(s) - T_d(s)$$

$$T_L(s) = J_m s^2 \theta(s) + B_m s \theta(s) \rightarrow \textcircled{6}$$

$$I_f(s) = \frac{V_f(s)}{R_f + L_f s}$$

$$\text{If } T_d = 0$$

$$T_L(s) = T_m(s)$$

$$[J_m s^2 + B_m s] \theta(s) = k_m I_f(s) = k_m \frac{V_f(s)}{R_f + L_f s}$$

$$\theta(s) / V_f(s) = \frac{k_m}{(R_f + L_f s) s (J_m s + B_m)}$$

